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#include <stdio.h>

int main() {
    int global_id;
    int m;
    int n;
    int semester;
    int course;
    int code;
    int credit;
    int GP;
    int semester_TC;
    int semester_TGC;
    int cumulative_TC;
    int cumulative_TGC;
    char grade;
    float SGPA;
    float CGPA;

    // Step 3 and 4: Get global_id and validate using while loop
    printf("Enter global_id: ");
    scanf("%d", &global_id);
    while (global_id <= 0) {
        printf("Invalid global_id! Please enter a positive value: ");
        scanf("%d", &global_id);
    }

    // Step 5 and 6: Get m and validate
    printf("Enter number of semesters: ");
    scanf("%d", &m);
    while (m <= 0) {
        printf("Invalid number of semesters! Please enter a positive value: ");
        scanf("%d", &m);
    }

    // Step 7: Set cumulative values
    cumulative_TC = 0;
    cumulative_TGC = 0;

    // Step 8: Loop for semesters
    for (semester = 1; semester <= m; semester = semester + 1) {
        printf("
Semester %d
", semester);

        // Step 8.1: Set semester totals
        semester_TC = 0;
        semester_TGC = 0;

        // Step 8.2 and 8.3: Get n and validate
        printf("Enter number of courses: ");
        scanf("%d", &n);
        while (n <= 0) {
            printf("Invalid number of courses! Please enter a positive value: ");
            scanf("%d", &n);
        }

        // Step 8.4: Loop for courses
        for (course = 1; course <= n; course = course + 1) {
            printf("

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Course %d
", course);
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// Step 8.4.1 and 8.4.2: Get code and validate
printf("Enter course code: ");
scanf("%d", &code);
while (code <= 0) {
    printf("Invalid code! Enter positive value: ");
    scanf("%d", &code);
}

// Step 8.4.3 and 8.4.4: Get credit and validate
printf("Enter credit: ");
scanf("%d", &credit);
while (credit <= 0) {
    printf("Invalid credit! Enter positive value: ");
    scanf("%d", &credit);
}
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// Step 8.4.5 and 8.4.6: Get grade
printf("Enter grade: ");
scanf(" %c", &grade);
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// Step 8.4.7: Find GP using if-else if
if (grade == 'O') {
    GP = 10;
} else if (grade == 'A') {
    GP = 9;
} else if (grade == 'B') {
    GP = 8;
} else if (grade == 'C') {
    GP = 7;
} else if (grade == 'D') {
    GP = 6;
} else if (grade == 'E') {
    GP = 5;
} else if (grade == 'P') {
    GP = 4;
} else {
    GP = 0;
}
```

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// Step 8.4.8: Calculate semester_TGC
semester_TGC = semester_TGC + (GP * credit);
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// Step 8.4.9: Calculate semester_TC
semester_TC = semester_TC + credit;
}
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// Step 8.5: Calculate SGPA
SGPA = (float)semester_TGC / semester_TC;
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// Step 8.6: Display semester and SGPA
printf("Semester: %d, SGPA: %f
", semester, SGPA);
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// Step 8.7 and 8.8: Add to cumulative totals
cumulative_TGC = cumulative_TGC + semester_TGC;
cumulative_TC = cumulative_TC + semester_TC;
}
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// Step 9: Calculate CGPA
CGPA = (float)cumulative_TGC / cumulative_TC;

// Step 10: Display final results
printf("
Global ID: %d
", global_id);
printf("Total Semesters: %d
", m);
printf("Cumulative Total Credits: %d
", cumulative_TC);
printf("CGPA: %f
", CGPA);

return 0;
}
```